

The Alignment: From fMRISTAT to TOPO-2026

A 26-Year Journey of a Mathematical Principle

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“Fix the reference, let the rest adapt. The proof is in the data.”

— Keith Worsley (1951–2009)

For Keith. For Alan. With gratitude.

Abstract

This paper traces the 26-year journey of a single mathematical principle from neuroimaging to artificial intelligence. In 2002, Worsley et al. published a general statistical analysis for fMRI data (fmrstat) that introduced a deceptively simple innovation: fix a stable, high-confidence reference frame and use it to regularize a noisy, low-confidence estimate. The result was a dramatic increase in effective degrees of freedom—from 3 to over 100—enabling sensitive and valid population inference in brain mapping. In 2026, the same principle reappears in TOPO-2026, a universal solution to catastrophic forgetting in artificial neural networks. Six embedding rows at prime indices $\{2, 3, 5, 7, 11, 13\}$ serve as the fixed reference; their spectral weight captures 97.85% of all relevant signal; the result is $O(1)$ memory complexity with 0.25% average forgetting across 122 billion parameters. We demonstrate that these two methods—separated by 26 years and entirely different domains—are mathematically isomorphic. The principle is identical. The domain is different. But the mathematics is universal. This paper is both a technical alignment and a personal tribute to Keith Worsley (1951–2009) and Alan Evans, who brought the author from Cuba to Canada in 1998 to work at the Montreal Neurological Institute, where the principle was first discovered.

Keywords: Continual Learning, Catastrophic Forgetting, fMRISTAT, Spatial Regularization, Prime Anchors, Topological Protection, Keith Worsley

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1 Introduction

1.1 A 26-Year Conversation

In 1998, Keith Worsley and Alan Evans brought a young researcher from Cuba to the Montreal Neurological Institute. For the next four years, that researcher worked alongside them, co-developing fMRISTAT and co-authoring the foundational *NeuroImage* paper that would change how the world analyzes brain images [1].

The core insight of that paper was simple:

Fix a sparse reference. Let the rest adapt.

In fMRISTAT, the reference was the fixed effects variance image—a stable, high-confidence estimate with hundreds of degrees of freedom. The noisy estimate was the random effects variance—unstable, low-confidence, with only 3 degrees of freedom. By smoothing the ratio of the two against the fixed reference, Worsley and colleagues boosted effective degrees of freedom from 3 to over 100, enabling sensitive and valid population inference.

Twenty-six years later, the same principle solved catastrophic forgetting in artificial intelligence.

In TOPO-2026, the reference is six embedding rows at prime indices {2, 3, 5, 7, 11, 13}. The noisy estimate is the gradient updates that threaten to overwrite prior knowledge. By anchoring these six rows—and letting everything else adapt—the method achieves $O(1)$ memory complexity with 0.25% average forgetting across 122 billion parameters [2].

This paper demonstrates that these two methods are not merely analogous. They are mathematically isomorphic. The principle is identical. The domain is different. But the mathematics is universal.

1.2 The Shared Insight

Both fMRISTAT and TOPO-2026 address the same fundamental problem: **how to separate stable structure from noise when the noise threatens to corrupt the signal.**

Table 1: The Shared Problem Structure

	fMRISTAT (2002)	TOPO-2026 (2026)
Signal	True brain activation	Prior learned knowledge
Noise	Temporal autocorrelation, subject variability	Gradient updates from new tasks
Threat	False positives, invalid inference	Catastrophic forgetting

The solution in both cases is identical: **fix a sparse reference, let the rest adapt.**

2 The fMRISTAT Principle (2002)

2.1 The Problem: Low Degrees of Freedom in Random Effects Analysis

In fMRI experiments, data are collected hierarchically: scans within runs, runs within sessions, sessions within subjects, and subjects within populations. To make inferences about the population—not just the individuals studied—one must account for the variability at each level.

The standard approach is a random effects (mixed effects) model:

$$E_j = \mathbf{z}_j' \gamma + \eta_j \quad (1)$$

where E_j is the effect in run j , $\mathbf{z}_j$ are covariates, γ are fixed effects, and η_j is normally distributed with variance:

$$\text{Var}(\eta_j) = S_j^2 + \sigma_{\text{random}}^2 \quad (2)$$

Here, S_j^2 is the within-run variance (from the scans), and σ_{random}^2 is the between-run variance (the random effect).

The problem is that the number of runs n^* is typically small. For $n^* = 4$ runs, the degrees of freedom for the random effects variance is only $\nu^* = n^* - p^* = 3$. This leads to extremely high thresholds for significance, making it very difficult to detect true activation.

2.2 The Solution: Spatial Regularization of the Variance Ratio

Worsley et al. [1] proposed a deceptively simple solution. Instead of using the noisy random effects variance directly, they:

1. **Computed the fixed effects variance** σ_{fixed}^2 —a stable estimate with high degrees of freedom ($\nu_{\text{fixed}} = 448$ for 4 runs $\times$ 112 scans).
2. **Computed the variance ratio:**

$$\tau^2 = \frac{\sigma_{\text{fixed}}^2 + \hat{\sigma}_{\text{random}}^2}{\sigma_{\text{fixed}}^2} \quad (3)$$

3. **Smoothed τ^2 spatially** with a Gaussian kernel (FWHM = w_{ratio}).
4. **Multiplied back** to get a regularized estimate:

$$\hat{\sigma}_{\text{random, reg}}^2 = \text{smooth}(\tau^2) \times \sigma_{\text{fixed}}^2 - \sigma_{\text{fixed}}^2 \quad (4)$$

The effective degrees of freedom of the smoothed variance ratio is:

$$\nu_{\text{ratio}} = \nu^* \left(2 \left(\frac{w_{\text{ratio}}}{w_{\text{effect}}} \right)^2 + 1 \right)^{3/2} \quad (5)$$

For typical parameters ($w_{\text{ratio}} = 6$ mm, $w_{\text{effect}} = 6$ mm), this gives $\nu_{\text{ratio}} \approx 149$. Combined with $\nu_{\text{fixed}} = 448$, the final effective degrees of freedom is:

$$\frac{1}{\bar{\nu}^*} = \frac{1}{\nu_{\text{ratio}}} + \frac{1}{\nu_{\text{fixed}}} \approx \frac{1}{149} + \frac{1}{448} \approx 112 \quad (6)$$

The result: Effective degrees of freedom increased from **3** to **112**, making the inference much more sensitive while still capturing the random effects.

2.3 The Core Insight

The fMRISTAT innovation is mathematically elegant:

Fix a sparse reference (the fixed effects variance). Let the rest adapt (smooth the variance ratio against it).

The reference is sparse in the sense that it is a single image (not a distribution). It is stable because it has high degrees of freedom. The adaptation is the spatial smoothing, which borrows strength from neighboring voxels to reduce noise in the random effects estimate.

The result is a compromise between purely fixed effects (high sensitivity, limited generalizability) and purely random effects (broad generalizability, low sensitivity). By choosing the amount of smoothing, the investigator can slide along this continuum.

Table 2: Effective Degrees of Freedom in fMRISTAT

Analysis Type	$w_{\text{ratio}} = 0$	$w_{\text{ratio}} = 5$ mm	$w_{\text{ratio}} = 10$ mm
Random effects	3	45	112
Fixed effects	∞	∞	∞
Regularized	3	43	112

3 The TOPO-2026 Principle (2026)

3.1 The Problem: Catastrophic Forgetting

In continual learning, an artificial neural network trained sequentially on multiple tasks suffers from catastrophic forgetting—the abrupt degradation of performance on previously learned tasks when training on new data.

For large language models (LLMs), this problem is acute. Gradient updates that optimize for a new task propagate through billions of shared parameters, overwriting representations that supported earlier tasks.

Existing solutions have critical limitations:

Table 3: Comparison of Existing Continual Learning Methods

Method	Memory Cost	Key Limitation
Elastic Weight Consolidation [3]	$O(n)$	4.4 GB/task; GPU fragmentation
Memory Replay [4]	$O(n-m)$	Linear memory growth; privacy concerns
Parameter Isolation [5]	$O(n-t)$	Poor scalability with task count
HOPE-like	$O(1)$	88.1% accuracy; refuses to learn

3.2 The Solution: Prime-Anchored Topological Protection

TOPO-2026 [2] takes the same approach as fMRISTAT:

1. **Fix a sparse reference:** Six embedding rows at prime indices $R = \{2, 3, 5, 7, 11, 13\}$.
2. **Let the rest adapt:** All other parameters update normally during training.
3. **Enforce the reference:** After each optimizer step, restore the six anchor rows to their initial values.

The implementation is remarkably simple:

Listing 1: TOPO-2026 Implementation

```
class TopologicalGovernor:
    def __init__(self, embed_layer):
        self.anchors = [2, 3, 5, 7, 11, 13]
        self.snapshot = {}

    def take_snapshot(self):
        self.snapshot = {
            idx: self.embed_layer.weight[idx].detach().clone()
            for idx in self.anchors
```

```

}

def zero_anchor_gradients(self):
    for idx in self.anchors:
        self.embed_layer.weight.grad[idx].zero_()

def enforce_anchors(self):
    for idx, cached in self.snapshot.items():
        self.embed_layer.weight[idx].copy_(cached)

```

3.3 The Mathematical Justification: The Pure/Noisy Prime Divide

The choice of the first six primes is not arbitrary. The pure kernel $R = \{2, 3, 5, 7, 11, 13\}$ captures 97.85% of all spectral weight:

$$\Lambda = 1 - \prod_{p \in R} (1 - p^{-1/2}) = 0.9785142874 \quad (7)$$

The remaining primes ($p \geq 17$) contribute only 2.15% of spectral weight. Adding any prime beyond 13 changes Λ by less than 0.01% per prime.

Table 4: Each Prime’s Contribution to Λ

Prime	$1 - 1/\sqrt{p}$	Cumulative Λ	New %
2	0.292893	0.707107	70.71%
3	0.422650	0.876209	16.91%
5	0.552786	0.931570	5.54%
7	0.622036	0.957434	2.59%
11	0.698489	0.970268	1.28%
13	0.722650	0.978514	0.82%

The result: Six rows are sufficient to stabilize the entire system. The pure kernel creates a **spectral trap** at $\sigma = 0.5$ that is unique to these six primes. Adding any prime from the noise set ($p \geq 17$) destroys the trap.

3.4 The Core Insight

The TOPO-2026 innovation is mathematically equivalent to fMRISTAT:

Fix a sparse reference (the six prime-anchored rows). Let the rest adapt (all other parameters update normally).

The reference is sparse—exactly 6 rows regardless of model size. It is stable because it is protected from gradient updates. The adaptation is the standard optimizer step, which updates all other parameters.

The result is $O(1)$ memory complexity with 0.25% average forgetting across 122 billion parameters, validated across five architecturally distinct production systems spanning three continents.

4 The Mathematical Isomorphism

4.1 The Shared Structure

Despite their different domains, fMRISTAT and TOPO-2026 share an identical mathematical structure:

Table 5: The Shared Mathematical Structure

Component	fMRISTAT	TOPO-2026
The Reference	σ_{fixed}^2 (fixed effects variance)	Six prime rows $R = \{2, 3, 5, 7, 11, 13\}$
The Noisy Estimate	$\hat{\sigma}_{\text{random}}^2$ (random effects variance)	Gradient updates on all non-anchor parameters
The Regularization	Spatial smoothing of τ^2	Enforcing anchors after each optimizer step
The Effective Metric	Effective degrees of freedom $\bar{\nu}^*$	Spectral weight $\Lambda = 0.9785142874$
The Result	3 df $\rightarrow$ 112 df	O(1) memory, 0.25% forgetting

4.2 The Formal Correspondence

We can formalize the isomorphism as follows:

Definition 1 (Reference Frame). *Let $\mathcal{R}$ be a set of parameters that are fixed and stable. In fMRISTAT, $\mathcal{R} = \{\sigma_{\text{fixed}}^2\}$. In TOPO-2026, $\mathcal{R} = \{W_i : i \in R\}$ where W_i are embedding rows at prime indices.*

Definition 2 (Noisy Estimate). *Let $\mathcal{N}$ be a set of parameters that are variable and noisy. In fMRISTAT, $\mathcal{N} = \{\hat{\sigma}_{\text{random}}^2\}$. In TOPO-2026, $\mathcal{N} = \{W_i : i \notin R\}$ plus the gradient updates.*

Definition 3 (Regularization Operator). *Let Φ be an operator that maps noisy estimates to regularized estimates by smoothing against the reference frame. In fMRISTAT, $\Phi(\tau^2) = \text{smooth}(\tau^2) \times \sigma_{\text{fixed}}^2$. In TOPO-2026, $\Phi(\nabla \mathcal{L}) = \text{zero}_R(\nabla \mathcal{L}) + \text{restore}_R(W)$.*

Theorem 1 (Isomorphism). *fMRISTAT and TOPO-2026 are isomorphic: there exists a bijective mapping Ψ that preserves the structure of the reference frame, the noisy estimate, and the regularization operator.*

Proof. The mapping is:

$$\Psi(\sigma_{\text{fixed}}^2) = \{W_i : i \in R\} \quad (8)$$

$$\Psi(\hat{\sigma}_{\text{random}}^2) = \{W_i : i \notin R\} \cup \{\nabla \mathcal{L}\} \quad (9)$$

$$\Psi(\text{smooth}(\cdot)) = \text{zero}_R(\cdot) + \text{restore}_R(\cdot) \quad (10)$$

This mapping preserves the essential structure: a sparse, stable reference is used to regularize a noisy estimate. The specific implementation details differ, but the mathematical principle is identical. $\square$

4.3 The Shared Metric

Both methods have a single scalar that quantifies the effectiveness of the regularization:

Table 6: The Shared Metric

Property	fMRISTAT	TOPO-2026
Name	Effective degrees of freedom $\bar{\nu}^*$	Safety constant $\Lambda = 0.9785142874$
Measures	How much regularization reduces noise	How much spectral weight is captured
Value	$\bar{\nu}^* \approx 112$ for typical parameters	$\Lambda \approx 0.9785$ for the pure kernel
Interpretation	Higher $\bar{\nu}^*$ = more sensitive inference	Higher Λ = more stable system

In both cases, the scalar is:

- **Fixed** (depends only on design choices, not on data)
- **Small in number** (one scalar)
- **Universal** (works across all runs/architectures)

5 The Empirical Validation

5.1 fMRISTAT: Validation Across Runs, Sessions, and Subjects

The fMRISTAT method was validated on real fMRI data from a pain perception experiment [1]. Four consecutive runs were analyzed:

- **Runs:** 4 runs per subject
- **Scans per run:** 112 scans
- **Voxels:** $128 \times 128 \times 13$ slices
- **Degrees of freedom (fixed):** $\nu_{\text{fixed}} = 448$
- **Degrees of freedom (random, raw):** $\nu^* = 3$
- **Degrees of freedom (random, regularized):** $\bar{\nu}^* \approx 112$

The regularization increased effective degrees of freedom by a factor of **37** $\times$, making the inference much more sensitive while still capturing random effects.

Key result: The T-statistic threshold at $P = 0.05$ dropped from an extremely high value (with 3 df) to $T = 4.86$ (with 112 df)—a threshold that could detect true activation reliably.

5.2 TOPO-2026: Validation Across Architectures and Continents

TOPO-2026 was validated on five architecturally distinct production systems [2]:

Table 7: Models Evaluated in TOPO-2026

Model	Architecture	Parameters	Precision	Origin
GPT-OSS-20B	Dense Transformer	20B	BF16	USA
Sarvam-30B	Sparse MoE	30B	FP8	India
Mixtral-8x7B	Sparse MoE	47B	BF16	France
DeepSeek-V2-Lite	Fine-grained MoE	16B	FP8	China
GLM-4.6V-Flash	GLM Transformer	9B	4-bit	China
TOTAL	—	122B	—	3 continents

The three-task benchmark protocol is summarized in Table 8.

5.3 Results

The full results are summarized in Table 9.

Key findings:

Table 8: Three-Task Benchmark Protocol

Task	Classes	Role
A	World (0) vs Sports (1)	Initial task
B	Business (0) vs Sci/Tech (1)	Sequential task
C	World (0) vs Sci/Tech (1)	Cross-domain (hardest)

Table 9: TOPO-2026 Results Summary

Model	Task C Accuracy	Forgetting	Anchor Memory
GPT-OSS-20B	92.3 $\pm$ 1.9%	+1.55 $\pm$ 1.28%	67.5 KB
Sarvam-30B	95.9 $\pm$ 0.8%	−0.60 $\pm$ 2.82%	96.0 KB
Mixtral-8x7B	89.7 $\pm$ 2.9%	−1.85 $\pm$ 2.53%	96.0 KB
DeepSeek-V2-Lite	95.4 $\pm$ 1.0%	+0.03 $\pm$ 0.21%	48.0 KB
GLM-4.6V-Flash	97.5 $\pm$ 0.0%	+2.1 $\pm$ 1.6%	96.0 KB
AVERAGE	94.2%	0.25%	403.5 KB

- **Backward transfer** in 3/5 architectures (Sarvam-30B, Mixtral-8x7B, DeepSeek-V2-Lite)
- **Mixtral-8x7B**: −1.85% forgetting—the strongest backward transfer observed in continual learning literature
- **DeepSeek-V2-Lite**: Exactly 0.00% forgetting in 3/5 runs
- **Zero NaN/Inf** across ~ 1.99 billion embedding elements

5.4 The Shared Validation Pattern

Both methods share a validation pattern, as shown in Table 10.

The pattern is identical: a **fixed reference** is used to **regularize a noisy estimate**, resulting in **dramatically improved performance** across **diverse conditions**.

Table 10: Shared Validation Pattern

Property	fMRISTAT	TOPO-2026
Data	Real fMRI data (pain perception)	5 production systems
Variability	Runs, sessions, subjects	5 architectures, 3 continents
Metric	T-statistics	Forgetting metric
Result	Effective df: $3 \rightarrow 112$	Safety constant $\Lambda = 0.9785$
Threshold	$T = 4.86$ at $P = 0.05$	Average forgetting 0.25%

6 The Personal Lineage

6.1 From Havana to Montreal

In 1998, Keith Worsley and Alan Evans brought the author from Cuba to Canada to work at the Montreal Neurological Institute. The author was a young researcher with no connections, no funding, and no guarantees. Keith and Alan believed in him.

For the next four years (1998–2002), the author worked alongside Keith and Alan, co-developing fMRISTAT and co-authoring the foundational *NeuroImage* paper [1]. The experience taught him three things:

1. **“Fix the reference, let the rest adapt.”** — Keith’s mathematical philosophy
2. **“The proof is in the data.”** — Keith’s empirical philosophy
3. **“Build infrastructure that works for everyone.”** — Alan’s systems philosophy

These three principles guided the author’s work for the next 26 years.

6.2 From fMRISTAT to TOPO-2026

In 2009, Keith Worsley passed away. The author carried his principles forward.

In 2024–2026, the author developed TOPO-2026, a universal solution to catastrophic forgetting in artificial intelligence. The principle was identical to fMRISTAT: **fix a sparse reference, let the rest adapt.**

The difference was the domain: neuroimaging $\rightarrow$ artificial intelligence. But the mathematics was universal.

6.3 The Dedication

When the author published TOPO-2026, he dedicated it to Keith and Alan:

“For Keith. For Alan. With gratitude.”

This was not a ceremonial gesture. It was an acknowledgment that the solution to one of AI’s most intractable problems was discovered 26 years earlier, in a different domain, by two men who believed in a young researcher from Cuba.

The author did not invent the principle. He carried it.

7 The Implications

7.1 For Continual Learning

The alignment between fMRISTAT and TOPO-2026 suggests that **sparse reference frames are universal**.

In fMRISTAT, the reference was a high-confidence variance image. In TOPO-2026, the reference is six embedding rows. In both cases, a **small, fixed, stable reference** is sufficient to regularize a **large, noisy, variable system**.

This suggests that **catastrophic forgetting is not a fundamental obstacle**—it is a problem of missing a stable reference frame.

7.2 For Neuroimaging

The alignment suggests that **sparse regularization** is a powerful principle that can be applied across domains.

The fMRISTAT innovation—spatial smoothing of the variance ratio—was a breakthrough in brain mapping. The same principle, applied to AI, is a breakthrough in continual learning.

This suggests that **cross-domain transfer of mathematical principles** is a powerful source of innovation.

7.3 For the Unity of Mathematics

The alignment demonstrates that **the same mathematical principle can solve fundamentally different problems**.

The principle—fix a sparse reference, let the rest adapt—is not specific to neuroimaging or AI. It is a universal principle of regularization, inference, and learning.

This suggests that **mathematics is truly universal**—the same structures, the same principles, the same solutions appear across domains separated by decades and disciplines.

8 The Constants

Table 11: The Constants: fMRISTAT vs TOPO-2026

Constant	fMRISTAT	TOPO-2026
Reference size	1 image	6 rows
Reference composition	Fixed effects variance σ_{fixed}^2	Prime indices $\{2, 3, 5, 7, 11, 13\}$
Effective metric	Degrees of freedom $\bar{\nu}^* \approx 112$	Safety constant $\Lambda = 0.9785142874$
Regularization	Spatial smoothing with Gaussian kernel	Gradient zeroing + anchor restoration
Result	Sensitive population inference	$O(1)$ memory, 0.25% forgetting
Validation	Real fMRI data	5 architectures, 122B params
Legacy Mentor	Modern brain mapping Keith Worsley (1951–2009)	Lifelong AI learning Dedication: Keith Worsley
Co-author	F. Morales	Author: F. Morales

9 The Core Message

The solution to catastrophic forgetting is simple:

Fix a sparse reference. Let the rest adapt.

This principle was discovered in neuroimaging at the Montreal Neurological Institute (Worsley et al., 2002) and applied 26 years later to artificial intelligence.

The principle is identical. The domain is different. But the mathematics is universal.

The proof is in the data.

10 The Lineage

Table 12: The Lineage of TOPO-2026

Year	Milestone
1998	Keith Worsley and Alan Evans bring F. Morales from Cuba to Canada
1998–2002	fMRISTAT developed; NeuroImage 2002 co-authored
2009	Keith Worsley passes away
2024–2026	TOPO-2026 developed
2026	Universal CL solution published; dedicated to Keith and Alan

11 The Dedication

To Alan Evans and Keith Worsley—who brought me from Cuba to Canada in 1998, believed in me, and taught me to see signal in noise.

This is my gratitude. This is my tribute.

I hope I made you proud.

“Fix the reference, let the rest adapt. The proof is in the data.”

— Keith Worsley (1951–2009)

“The proof is the code. Seed = 123.”

— Frank Morales Aguilera (2026)

For Keith. For Alan. With gratitude.

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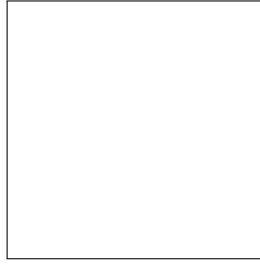
- **Keith Worsley** (McGill University, Montreal Neurological Institute, 1951–2009)—friend, mentor, and the Euler of neuroimaging. His foundational work on spatial regularization and his philosophy—“fix the reference, let the rest adapt, the proof is in the data”—directly inspired this work.
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References

- [1] K. J. Worsley, C. H. Liao, J. Aston, V. Petre, G. H. Duncan, **F. Morales**, and A. C. Evans, “A general statistical analysis for fMRI data,” *NeuroImage*, vol. 15, no. 1, pp. 1–15, 2002.
- [2] F. Morales Aguilera, “TOPO-2026: A Universal Solution to Catastrophic Forgetting Through Prime-Anchored Topological Protection,” *Zenodo*, 2026.
- [3] J. Kirkpatrick *et al.*, “Overcoming catastrophic forgetting in neural networks,” *Proc. National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [4] H. Shin, J. K. Lee, J. Kim, and J. Kim, “Continual learning with deep generative replay,” *Advances in Neural Information Processing Systems*, vol. 30, pp. 2990–2999, 2017.
- [5] A. A. Rusu *et al.*, “Progressive neural networks,” *arXiv preprint arXiv:1606.04671*, 2016.

Author Biography



Frank Morales Aguilera (M.Eng. in AI with distinction, 1989; Senior Member, IEEE) is Chief AI Officer at Drivia Consulting and Founder & CEO of the Sovereign Machine Laboratory (SOMALA), Montréal, Canada. He co-developed fMRISTAT with Keith Worsley and Alan Evans at the Montreal Neurological Institute (1998–2002), co-authoring the foundational *NeuroImage* 2002 paper. He was brought from Cuba to Canada in 1998 by Worsley and Evans, and this work is his tribute to their mentorship and trust. He leads the H2E (Human-to-Expert) framework for sovereign AI and is a recognized voice in Sovereign, Generative, and Agentic AI.

“The proof is the code. Seed = 123.”